

Capital Investment Assumptions		
Metric	Option A	Option B
Initial Investment	\$2,800,000	\$2,650,000
Projected Annual Revenue	\$900,000	\$1,625,000
Operating Margin	21.0%	25.0%
Estimated 1-Year VaR	\$220,000	\$360,000
Market Risk Level	Low	High
Inflation Assumption	4.50%	4.50%
Note: These are illustrative sample assumptions generated for this project and are not sourced from market data.		
Calculating Annual Operating Profit		
Calculation	Option A	Option B
Formula	Revenue × Operating Margin	Revenue × Operating Margin
Annual Operating Profit (\$)	\$189,000	\$406,250
Inflation Risk Consideration		
Long-term inflation can raise labor, materials, energy, and financing costs. If project revenues cannot be repriced at the same pace, operating margins and real returns may decline. This workbook uses an illustrative inflation assumption of 4.5% for risk discussion.		

Calculated Annual Operating Profit

Calculation	Option A	Option B
Formula	Revenue × Operating Margin	Revenue × Operating Margin
Annual Operating Profit (\$)	\$189,000	\$406,250

Investment Comparison

Metric	Option A	Option B
Initial Investment	\$2,800,000	\$2,650,000
Projected Annual Revenue	\$900,000	\$1,625,000
Operating Margin	21.0%	25.0%
Estimated Annual Operating Profit	\$189,000	\$406,250
Estimated 1-Year VaR	\$220,000	\$360,000
Market Risk Level	Low	High

Risk Matrix Summary

Risk Category	Option A	Justification
Financial Risk	High	Annual operating profit of \$189,000 is relatively low compared with the \$2.8 million initial investment, while the estimated VaR of \$220,000 exceeds annual operating profit. This creates a longer recovery period and greater downside exposure.
Market Risk	Low	Option A has lower projected revenue and appears to rely on more conservative market expectations. Its lower VaR also indicates less exposure to market volatility.
Operational Risk	Moderate	The smaller revenue target and lower operating margin suggest a relatively manageable operation, although inflation and unexpected cost increases could place pressure on its 21% margin.

Risk Category	Option B	Justification
Financial Risk	Moderate	Although Option B has a higher VaR of \$360,000, its stronger annual operating profit of \$406,250 and lower initial investment of \$2.65 million provide greater capacity to absorb financial losses.
Market Risk	High	Option B depends on significantly higher revenue of \$1.625 million, making its performance more sensitive to changes in customer demand, competition, pricing, and economic conditions.
Operational Risk	High	Achieving the higher revenue and 25% margin may require greater production capacity, staffing, supply-chain coordination, and cost control, increasing execution risk.

To: Chief Financial Officer

Re: Capital Investment Recommendation – Option A vs. Option B

Following the financial and risk assessment of the two proposed capital investment alternatives, I recommend that the company proceed with Option B, subject to appropriate risk controls and periodic performance reviews.

From a financial return perspective, Option B provides the stronger investment case. It requires an initial investment of approximately \$2.65 million, compared with \$2.80 million for Option A, yet generates significantly higher projected annual revenue of \$1.625 million versus \$900,000. With an operating margin of 25%, Option B is expected to deliver annual operating profit of approximately \$406,250, more than double Option A's projected profit of \$189,000. This indicates superior capital efficiency and stronger potential to support profitability and return-on-investment targets.

The higher return, however, comes with increased risk exposure. Option B has an estimated one-year Value at Risk of \$360,000, compared with \$220,000 for Option A, and carries higher market and operational risk. Its performance depends more heavily on achieving forecast demand, maintaining pricing power, controlling operating costs, and executing effectively at a larger scale. Nevertheless, its projected operating profit provides a greater financial cushion against adverse performance than Option A.

Current economic considerations also reinforce the need for a risk-adjusted approach. The project assumes inflation of approximately 4.5%, and sustained increases in labour, energy, materials, financing, and supply-chain costs could place pressure on operating margins. Option B's stronger 25% margin provides better capacity to absorb inflationary pressures, although management should closely monitor cost escalation and market demand.

Option B is also better aligned with key company KPIs, particularly revenue growth, operating profit, operating margin, capital efficiency, and risk-adjusted return. Management should establish quarterly performance thresholds for revenue, margin, operating costs, and VaR to ensure that actual results remain within acceptable limits.

Based on the superior revenue potential, profitability, lower initial capital requirement, and stronger ability to absorb economic pressures, Option B should be approved as the preferred investment, with disciplined cost management and ongoing risk monitoring forming part of the implementation plan.

KPI & Economic Signal Explanation

Rising interest rates could affect both investment options by increasing the company's cost of borrowing and required rate of return. Because Option A requires a higher initial investment of \$2.8 million, higher financing costs could further weaken its already lower annual operating profit of \$189,000.

Option B requires slightly less capital at \$2.65 million and generates a stronger operating profit of \$406,250, giving it greater capacity to absorb increased interest expenses. However, Option B's higher market exposure means rising rates could also reduce customer demand and increase its overall risk.

The key KPIs affected by the investment decision include annual revenue, operating profit, operating margin, return on investment, payback period, cash flow, Value at Risk (VaR), and capital efficiency. Option B performs more strongly on revenue, profitability, margin, and capital efficiency, although its higher VaR requires closer risk monitoring.

Economic conditions are important in capital budgeting because factors such as inflation, interest rates, economic growth, consumer demand, and financing availability directly influence project costs, revenues, cash flows, and investment returns.

A project that appears profitable under favourable conditions may become less attractive when borrowing costs or operating expenses increase. Therefore, capital investment decisions should evaluate both expected financial returns and the company's ability to withstand adverse economic conditions.